pressure on the prey species is to escape predation and survive, so as to have more offspring. Meanwhile, the predator is under pressure to have a higher predation rate in order to provide food for more offspring. However, neither species is superior, responding instead to the adaptations of the other. The predator–prey relationship between even-toed hoofed mammals —such as antelopes and deer—and mammalian carnivores, like the big cats and wolves, is an example of this evolutionary arms race. The hoofed animals have long legs, extended by walking on the very tips of thickened and fused toe bones. This adaptation allows them to outrun and outjump their predators. In response, big cats—such as lions and tigers—have evolved speed and strength to bring down large, fleet-footed prey in surprise attacks. Wolves have evolved the stamina to run for long distances without stopping. This allows them to work as a team to chase down their prey and kill them when the exhausted prey collapse.

While the predator–prey equations offer an insight into the population dynamics of two species, the assumptions they rely on are rarely reflected in real life. Some predators do specialize in killing a single prey species, but other factors in the ecosystem also affect their populations.

Other applications

The Lotka–Volterra equations have been used to study the dynamics of food chains and food webs in which one species may be a predator of another species but also the prey species of a third. They have also been used to examine the relationship between host and parasite species, which bears some resemblance to that between prey and predator. Parasites often specialize in one host species—a relationship that should resemble the one described by the Lotka–Volterra equations. However, in practice the process of evolution is thought to interfere with this. A parasite does not usually kill its host (those that do are called parasitoids), but can reduce its fitness. The Red Queen evolutionary theory, proposed in the 1970s by Leigh Van Valen,